

RIVERTON, WY, USA

WMO: 726720

Lat: 43.0647N

Lon: 108.4767W

Elev: 1699

StdP: 82.51

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 24061

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.2	-19.1	-25.2	0.5	-21.0	-22.5	0.6	-17.6	14.3	2.4	12.4	-0.3	1.5	330	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	33.6	14.6	32.3	14.3	30.9	14.1	16.5	27.8	15.8	27.3	15.2	26.8	5.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	11.5	18.6	12.1	10.8	18.5	10.9	10.0	18.6	52.2	27.8	49.9	27.2	47.8	26.7	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	11.0	9.1	-27.0	36.3	3.5	1.3	-29.5	37.2	-31.6	37.9	-33.5	38.7	-36.1	39.6
			-27.1	17.9	3.2	0.9	-29.4	18.5	-31.3	19.0	-33.2	19.5	-35.5	20.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-6.4	-3.9	2.3	6.7	11.9	17.8	22.9	21.2	15.5	7.5	0.0	-6.5	(6)
(7)		DBStd	11.32	6.34	6.32	5.61	4.75	4.79	4.18	2.78	3.01	4.73	5.39	6.19	6.29	(7)
(8)		HDD10.0	2229	509	389	241	119	37	2	0	0	11	108	302	511	(8)
(9)		HDD18.3	4276	767	622	496	350	205	58	3	11	107	336	551	769	(9)
(10)		CDD10.0	1306	0	0	3	19	94	236	399	346	176	31	1	0	(10)
(11)		CDD18.3	311	0	0	0	0	4	42	144	99	21	0	0	0	(11)
(12)		CDH23.3	4061	0	0	0	6	103	643	1770	1208	321	11	0	0	(12)
(13)		CDH26.7	1670	0	0	0	1	19	244	826	490	90	1	0	0	(13)
(14)	Wind	WSAvg	3.7	3.1	3.4	4.1	4.5	4.1	3.9	3.8	3.7	3.7	3.3	3.0		(14)
(15)	Precipitation	PrecAvg	208	6	7	11	26	40	34	18	12	20	19	11	8	(15)
(16)		PrecMax	343	37	34	75	94	113	151	74	40	113	91	54	91	(16)
(17)		PrecMin	95	0	0	0	2	3	0	1	0	0	0	0	0	(17)
(18)		PrecStd	59	7	7	13	19	29	31	16	10	21	19	11	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	12.7	20.0	23.8	28.8	34.1	36.3	34.2	31.8	24.9	17.4	11.2	(19)
(20)			MCWB	2.8	4.4	7.2	9.2	12.2	13.7	15.0	14.8	14.3	10.7	7.0	3.8	(20)
(21)		2%	DB	7.1	9.7	16.6	21.0	26.3	32.0	34.1	32.6	29.0	21.6	13.8	7.4	(21)
(22)			MCWB	1.7	2.8	5.8	8.0	11.3	13.7	14.8	14.3	12.7	9.6	5.5	1.8	(22)
(23)		5%	DB	4.6	7.3	13.9	18.1	23.7	29.5	32.8	31.3	27.3	18.7	11.3	4.7	(23)
(24)			MCWB	0.2	1.4	4.8	7.1	10.8	13.2	14.7	14.1	12.3	8.6	4.3	0.3	(24)
(25)		10%	DB	2.5	4.7	11.3	16.0	21.3	27.3	31.3	29.7	24.7	16.3	8.7	2.3	(25)
(26)			MCWB	-1.1	0.0	3.7	6.2	10.0	12.7	14.4	13.6	11.5	7.4	3.0	-1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	5.1	7.7	10.1	14.1	16.1	17.8	16.9	15.7	11.8	7.5	4.1	(27)
(28)			MCDB	8.1	11.9	18.1	21.6	25.6	28.0	28.2	28.7	26.4	22.5	15.5	10.0	(28)
(29)		2%	WB	1.9	3.3	6.3	8.6	12.4	15.1	16.9	16.1	14.3	10.3	5.8	2.2	(29)
(30)			MCDB	6.4	8.9	15.4	18.8	22.9	27.2	27.8	27.6	25.4	19.5	13.1	7.2	(30)
(31)		5%	WB	0.4	1.7	5.2	7.7	11.5	14.3	16.2	15.4	13.3	9.1	4.5	0.5	(31)
(32)			MCDB	4.4	6.8	13.3	17.2	21.7	26.1	27.5	26.7	23.8	17.6	10.8	4.5	(32)
(33)		10%	WB	-1.0	0.1	3.9	6.6	10.6	13.5	15.5	14.8	12.4	8.0	3.0	-1.1	(33)
(34)			MCDB	2.3	4.4	10.8	15.0	19.9	24.7	27.2	26.2	22.6	15.6	8.3	2.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.5	11.2	12.8	13.1	13.6	15.3	16.2	15.7	14.7	12.8	11.5	10.3	(35)
(36)			MCDBR	12.4	13.7	16.1	17.2	17.3	18.6	17.9	17.6	16.0	14.7	12.8		(36)
(37)			MCWBR	8.4	8.4	8.4	8.0	6.8	6.6	6.1	6.4	7.0	7.6	8.4	8.5	(37)
(38)		5% WB	MCDBR	12.1	13.5	15.4	15.9	15.9	16.7	15.9	15.8	15.6	15.2	14.3	12.5	(38)
(39)			MCWBR	8.4	8.5	8.1	7.6	6.4	6.4	5.4	5.7	6.3	7.5	8.4	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.233	0.239	0.258	0.297	0.312	0.314	0.339	0.331	0.302	0.262	0.242	0.233	(40)	
(41)		taud	2.557	2.560	2.537	2.435	2.456	2.472	2.429	2.437	2.512	2.612	2.583	2.548	(41)	
(42)		Ebn at Noon	943	989	999	972	960	954	926	925	934	946	923	911	(42)	
(43)		Edh at Noon	64	77	90	108	109	108	112	107	92	72	62	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.09	3.12	4.42	5.51	6.37	7.53	7.30	6.34	5.13	3.47	2.32	1.70	(44)	
(45)		RadStd	0.14	0.19	0.24	0.32	0.54	0.45	0.29	0.20	0.34	0.23	0.17	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page